Venkatesh Parvathala

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.15
Rajiv Gandhi University of Knowledge Technologies, RK Valley, Andhra Pradesh	Class XII (Rajiv Gandhi University of Knowledge Technologies, Andhra Pradesh)	2017	9.59%
Z P High School Kanduru, Chittoor Dist.	Class X (Secondary School Certificate, Andhra Pradesh)	2015	9.80%

PUBLICATIONS

Parvathala V, Andhavarapu S, Pamisetty G, Murty K. Neural Comb Filtering using Sliding Window Attention Network for Speech Enhancement. Circuits, Systems, and Signal Processing (pp. 1-22)

Parvathala V, Murty K. Light-Weight Causal Speech Enhancement Using Time-Varying Multi-Resolution Filtering. In Proc. National Conference on Communications (NCC), Chennai, India (pp. 1-6). doi: 10.1109/NCC60321.2024.10485757

Parvathala V, Gundluru R, Sankala S, Murty K. Exploiting Bispectral Features for Single-Channel Speech Enhancement. In Proc. Interspeech 2025 (pp. 2385-2389)

Behera A, Easow RA, **Parvathala V**, Murty K. Test-Time Training for Speech Enhancement. In Proc. Interspeech 2025 (pp. 2375-2379)

Sankala S, **Parvathala V**, Gundluru R, Murty K. Adversarial Attacks on Text-dependent Speaker Verification System. In Proc. Interspeech 2025 (pp. 4558-4562)

Parvathala V, Murty K. MSFNet: A Nested Model for Multi-Sampling-Frequency Speech Enhancement. In Proc. Interspeech 2025 (pp. 5138-5142)

Parvathala V, Murty K. Dynamic Layer Gating for Speech Enhancement. In Proc. Interspeech 2025 (pp. 2365-2369)

EXPERIENCE

Internship at Microsoft - Munich, Germany

Worked on real-time personalized speech enhancement for conversational speech.

Internship at ai-coustics - Berlin, Germany

Worked on real-time speech enhancement models for machine perception and interaction.

SKILLS

Python, Pytorch, Tensorflow, Git

RELEVANT COURSES

Pattern Recognition and Machine Learning

Convex Optimization

Numerical Linear Algebra

Random Variables and Stochastic Processes

Information Theory

Advanced Digital Signal Processing

Speech Systems

PROJECTS

Single-Channel Speech Enhancement

Developed a causal, light-weight speech enhancement method using a production-related loss function, deployed in automotive applications.

Multi-Channel Speech Enhancement

Created an end-to-end deep neural network for denoising and dereverberation in multi-channel car environments.

Personalized Speech Enhancement

Adapted light-weight models for target speaker separation, enhancing performance with a pitch-based loss function.

Speech-to-Speech Translation

Investigated cascading and Translatotron-based models for translating Indian languages.

Machine Translation

Explored attention-based and transformer architectures for English-to-Hindi translation, supporting speech-to-speech translation systems.

OTHER ACTIVITIES

Chaired a technical session at **Interspeech 2025**

Developed course content for the **ICASSP 2025** education short course on "**Phase Processing of Speech Signals**"

Serving as an active reviewer for **Interspeech** (international conference), **Computer Speech & Language** (international journal), and **APSIPA Transactions on Signal and Information Processing** (international journal)

Served as a teaching assistant for **signal processing**, **machine learning**, and s**peech processing** courses at IIT Hyderabad

Attended a 10-day course on "**Mathematical Foundations for Array Signal Processing and its Applications**" by **Israel Cohen** (Dec 2024)

Attended HAHT-IUS 2024, an Indo-US workshop on **"Harnessing AI for Healthcare Transformation"**

Represented IIT Hyderabad at the **Inter-IIT 2024** squash competitions